



An adaptive remaining useful life prediction approach for single battery with unlabeled small sample data and parameter uncertainty

Jiushi Zhang^a, Yuchen Jiang^a, Xiang Li^a, Mingyi Huo^a, Hao Luo^{a,*}, Shen Yin^b

^a Department of Control Science and Engineering, School of Astronautics, Harbin Institute of Technology, Harbin, China

^b Department of Mechanical and Industrial Engineering, Faculty of Engineering, Norwegian University of Science and Technology, Trondheim 7034, Norway

ARTICLE INFO

Keywords:

Battery
Remaining useful life
Unlabeled small sample data
Parameter uncertainty
KF-EM-RTS
Prediction

ABSTRACT

Accurate prediction of the remaining useful life (RUL) of lithium-ion battery is of great significance for the reliability of electronic equipment. In the conventional approaches, there are notable challenges in the RUL prediction for a single battery lacking historical data. To predict the battery's RUL under the condition of unlabeled small sample data and to describe the uncertainty of the parameter estimation in the degradation model, a novel adaptive approach based on Kalman filter and expectation maximum with Rauch-Tung-Striebel (KF-EM-RTS) is proposed to predict the battery's RUL. Specifically, without RUL labels and offline training, an online KF adaptive-update model based on the Wiener process is proposed for a single battery, in which the uncertainty of parameter estimation is described. Furthermore, the unknown model parameters can be adaptively estimated using EM-RTS to overcome the constraints of strong Markov characteristics, the convergence of which is proved. The real-world battery dataset provided by NASA Ames research center is applied to verify the proposed RUL prediction approach. Experimental results show that the proposed approach outperforms the existing conventional data-driven approaches for predicting the battery's RUL.

1. Introduction

1.1. Research background and related work

Lithium-ion batteries have widespread applications in machinery manufacturing, vehicle engineering, aerospace and other fields due to many advantages, including low self-discharge rate, high energy efficiency, long charge and discharge cycle, and stable electrochemical characteristics [1–4].

In general, the degradation of the battery can be categorized into calendar aging and operational aging. Calendar aging is always taking place, no matter whether it is sitting idle or working. By contrast, operational aging occurs when the battery is subjected to external factors, including: over-charging, over-discharging, high-temperature, and over-current [5,6]. In this sense, the capacity of the battery gradually decreases, which has a serious influence on the reliability and safety of the electronic equipment [7]. Therefore, accurate acquisition of the degradation information is of great significance for batteries. Remaining useful life (RUL) is an indispensable indicator to measure the degradation process of a battery, which refers to the time interval from the current state to the failure of battery for the first time [8,9]. Accurate RUL prediction of battery can provide good power performance and promote the reliability of the battery. Furthermore, the

operation of the battery can be monitored in time, which contributes to the reduction of the maintenance cost [10–12].

At present, the RUL prediction approaches for batteries can be generally divided into physical-of-failure approaches, data-driven approaches, and hybrid empirical approaches [13,14]. The physical-of-failure RUL prediction approaches are to analyze the operating mechanism of battery. Based on the physical laws, the corresponding equivalent circuit model is constructed to predict the RUL. Nevertheless, due to the complexity in internal operating mechanism and uncertain degradation condition, there are difficulties in the battery mechanism modeling processes [15]. On the other hand, without sufficient prior knowledge, an algorithm model can be constructed through data-driven approaches to predict battery's RUL. According to the types of algorithm, data-driven approaches can be categorized into machine learning-based and deep learning-based approaches. For machine learning-based approaches, Gao et al. used a new multi-core support vector machine optimized by particle swarm optimization (MCSVM-PSO) to predict the battery's RUL [16]. Different from machine learning approaches, deep learning approaches, such as convolutional neural network (CNN) [17], and long short-term memory network (LSTM) [18] predict RUL with the aid of deep neural networks. Wang et al. conducted a comprehensive review of the works in recent years using improved deep

* Corresponding author.

E-mail address: hao.luo@hit.edu.cn (H. Luo).

<https://doi.org/10.1016/j.ress.2022.108357>

Received 16 December 2021; Received in revised form 21 January 2022; Accepted 23 January 2022

Available online 19 February 2022

0951-8320/© 2022 Elsevier Ltd. All rights reserved.

Table 1

A summary of the comparative bibliographical studies.

Year	Approach type	Battery type/Dataset	Approach name
2017	Data-driven (machine learning)	18 650 Lithium-ion battery	MCSVM-PSO [16]
2020	Hybrid empirical approach	18 650 Lithium-ion battery	LSTM [20]
2021		18 650 Lithium-ion battery	GRU [21]
2021		CALCE dataset	PF-SVR [24]
2020		18 650 Lithium-ion battery	UKF-SVR [25]
2018		18 650 Lithium-ion battery	LS-EKF [26]
2018	Hybrid empirical approach	LiMn ₂ O ₄ battery	GH-PF [27]
2020		LiCoMn ₂ NiO ₂ battery	ICS-PF [28]
2021		Equivalent circuit model	ANFIS and SVR [29]

learning methods for battery RUL prediction, and proposed a deep convolutional neural network-extreme learning machine algorithm to accurately predict battery's RUL [19]. Considering multiple measurable data from the battery management system, Park et al. established an LSTM network to predict RUL under the phenomenon of battery capacity regeneration [20]. Gated recurrent unit (GRU) is an improved LSTM, Ardeshtari et al. proposed GRU trained by adaptive gradient descent algorithm to predict the RUL under nonlinear capacity degradation [21]. Data-driven approaches have widespread applications in the field of battery RUL prediction. Conversely, these learning approaches are essentially black-box models, which require massive offline data for training algorithm [22,23]. Furthermore, the uncertainty in the parameter estimation in the battery degradation process cannot be described.

The hybrid empirical approaches integrate the prior knowledge to the algorithm model. In detail, a state update model with state space variables is constructed. Furthermore, the algorithms such as particle filter (PF) [24], unscented Kalman filter (UKF) [25], and extended Kalman filter (EKF) [26] are applied for parameter identification, so as to accomplish the RUL prediction, which have strong explainability and reliability. In order to improve the conventional PF algorithm, Ma et al. developed a Gauss-Hermite-PF (GH-PF), which can predict the RUL of battery under nonlinear and non-Gaussian conditions [27]. Combining KF and PF algorithm, Qiu et al. discussed a multi-scale hybrid KF algorithm to estimate the state-of-charge (SOC) of the battery, the RUL of which is predicted based on an improved cuckoo search PF algorithm (ICS-PF) [28]. Xue et al. combined adaptive unscented KF (AUKF) and genetic algorithm-support vector regression (GA-SVR) algorithm to predict the RUL of the battery [25]. As for the integration of the learning-based time series prediction and model-based prognosis approach, Aggab et al. proposed an adaptive neuro-fuzzy inference system (ANFIS) and the support vector regression (SVR) for battery's RUL prediction. Experimental results show that the proposed approach has great performance [29]. The comparative bibliographical studies in this paper are summarized in Table 1, where CALCE is the abbreviation of Center for Advanced Life Cycle Engineering at the University of Maryland.

1.2. Motivations and contributions of this work

Most of the data-driven RUL prediction approaches need massive historical data and the corresponding labeled RUL to train models offline. Alternatively, the online data are required to follow a similar distribution with the training offline data in the model [30,31]. However, there are notable limitations to obtain historical degradation data of multiple batteries of the same type in practice, especially for the newly installed batteries in critical systems [23,32].

Inspired by related work, the motivations of this work can be summarized in the following aspects:

- Without historical data of multiple batteries of the same type, there are challenges in the RUL prediction through the online degradation data of a single battery [23,33].

- Conventional data-driven approaches require substantial historical data and RUL labels. In this sense, it is an essential challenge to predict the RUL under the condition of unlabeled small sample [32,34].
- The uncertainty of parameter estimation in battery degradation models is difficult to quantify, which requires in-depth research [8,33].
- Some existing prediction approaches are restricted by strong Markov characteristics [35]. Taking advantage of the data until the current time of the battery in an adaptive manner for predicting RUL needs to be further studied.

Based on these observations, a novel adaptive approach based on Kalman filter and expectation maximum with Rauch-Tung-Striebel (KF-EM-RTS) is proposed to predict the battery's RUL. The main contributions can be summarized in three folds:

- Without any historical data for offline learning, only the capacity data are needed to accomplish the RUL prediction for a single battery under the situation of unlabeled small sample data, from which the uncertainty of the parameter estimation in the degradation model is described by a probability density function.
- Based on the capacity data until the current time of the battery, EM-RTS is proposed to adaptively estimate the unknown parameters in the dynamic degradation model, which can overcome the constraints of strong Markov characteristics. The convergence of the algorithm is proved.
- The battery dataset provided by NASA is applied to verify the proposed RUL adaptive prediction approach. Experimental results show that the proposed approach outperforms the existing approaches for predicting the battery's RUL.

The remainder of this paper is organized as follows: Section 2 formulates the research problem. Section 3 presents the proposed adaptive RUL approach. Section 4 analyzes the battery dataset provided by NASA Ames research center. Section 5 summarizes this paper.

2. Problem formulation

Wiener process (WP) is widely applied in equipment degradation modeling, the essence of which is a diffusion process with a linear drift coefficient following a Brownian motion [35,36]. Brownian motion is used to describe the random motion of tiny particles, the mean value of which is 0, and the variance of which is subordinated to Gaussian process related to time [35]. In this paper, the battery capacity is used to describe the battery degradation process. The WP $\{q(t), t \geq 0\}$ can be modeled as:

$$q(t) = \alpha t + \beta B(t) \quad (1)$$

where $q(t)$ is the battery capacity at the time t , α is the drift coefficient related to the degradation process, which is equal to the degradation rate of the capacity. β is the variation coefficient, which characterizes the degree of uncertainty in the degradation process. $B(t)$ is the Brownian motion that satisfies $B(t) \sim N(0, t)$. In this paper, the research objective is to construct a dynamic degradation model based on the WP to predict RUL for a single battery, facing the problems of unlabeled small sample data and uncertainty of the parameter estimation in the battery degradation model. The single battery studied in this paper refers to only one battery rather than a batch of batteries.

It should be noted that the state of charge (SOC) is a key indicator to measure the remaining capacity of the battery. The United States Advanced Battery Consortium (USABC) defines SOC of the battery as the ratio between the present capacity q_p and the rated capacity q_r :

$$SOC = \frac{q_p}{q_r} \times 100\% \quad (2)$$

The articles [32,37,38] discussed that the performance of battery is unreliable to meet the requirements of the devices, when the SOC is

lower than 80%. At this time, maintenance or replacement actions for the battery are required. On this basis, this paper defines the RUL of the battery as the time interval between the current moment and the time when the SOC drops to 80% for the first time.

3. The methodology of RUL prediction

3.1. Adaptive state update model based on Kalman filter

Combined with Eq. (1) of the WP in Section 2, the battery capacity degradation process can be specifically shown in Eqs. (3) and (4):

$$\alpha_k = \alpha_{k-1} + \varphi \quad (3)$$

$$q_k = q_{k-1} + \alpha_{k-1} (t_k - t_{k-1}) + \beta \xi_k \quad (4)$$

where $\varphi \sim N(0, \sigma_\varphi^2)$, $\xi_k \sim N(0, t_k - t_{k-1})$. Considering that the time-varying battery capacity degradation rate in the current cycle depends on the previous time, φ is used to describe the uncertainty of α . Furthermore, $\beta \xi_k$ is used to describe the drift of the standard Brownian motion. On this basis, Eqs. (3) and (4) can be reconstructed by the dynamic state space model in Eq. (5) to describe the battery degradation process.

$$\begin{cases} x_k = A_k x_{k-1} + w_k \\ y_k = C_k x_k + e_k \end{cases} \quad (5)$$

where $x_k \in R^{2 \times 1}$, $A_k \in R^{2 \times 2}$, $w_k \in R^{2 \times 1}$, $C_k \in R^{1 \times 2}$. Specifically, it can be described in Eq. (6).

$$\begin{cases} x_k = \begin{bmatrix} \alpha_k \\ q_k \end{bmatrix}, A_k = \begin{bmatrix} 1 & 0 \\ t_k - t_{k-1} & 1 \end{bmatrix}, w_k = \begin{bmatrix} \varphi \\ \beta \xi_k \end{bmatrix} \\ C_k = [0 \quad 1], e_k = \nu \end{cases} \quad (6)$$

where α_k is the battery capacity degradation rate under the k th working cycle, q_k is the corresponding battery capacity, w_k and e_k are the noises of the state variables and the measurement equation, respectively, ν is the measured noise that can be expressed as $\nu \sim N(0, \sigma_\nu^2)$, y_k is the measured battery capacity under the k th working cycle, the covariance matrix of w_k can be expressed as $Q_k = \begin{bmatrix} \sigma_\varphi^2 & 0 \\ 0 & \beta^2 (t_k - t_{k-1}) \end{bmatrix}$.

In summary, the dynamic state space model consists of observable measurement vector $y_{1:k} = [y_1, y_2, \dots, y_k]$, state space vector $x_{1:k} = [x_1, x_2, \dots, x_k]$ and state space model parameters $\Xi = [A_k, w_k, C_k, e_k]$. According to $y_{1:k}$, the estimated posterior expectation of state matrix $\hat{x}_{k|k}$ and covariance matrix $S_{k|k}$ are shown in Eqs. (7)–(8):

$$\hat{x}_{k|k} = \begin{bmatrix} \hat{\alpha}_{k|k} \\ \hat{q}_{k|k} \end{bmatrix} = E(x_k | y_{1:k}) = \begin{bmatrix} E(\alpha_k | y_{1:k}) \\ E(q_k | y_{1:k}) \end{bmatrix} \quad (7)$$

where $\hat{\alpha}_k$ and $\hat{q}_k$ are the estimated posterior expectation of the two state variables, respectively.

$$S_{k|k} = cov(x_k | y_{1:k}) = \begin{bmatrix} var(\alpha_k | y_{1:k}) & cov(\alpha_k, q_k | y_{1:k}) \\ cov(\alpha_k, q_k | y_{1:k}) & var(q_k | y_{1:k}) \end{bmatrix} \quad (8)$$

where $cov(\alpha_k, q_k | y_{1:k})$ is the covariance of the two state variables, $var(\alpha_k | y_{1:k})$ and $var(q_k | y_{1:k})$ are the variance of the two state variables, respectively. Correspondingly, the one step forward prediction of state matrix $\hat{x}_{k|k-1}$ and covariance matrix $S_{k|k-1}$ are shown in Eqs. (9)–(10):

$$\hat{x}_{k|k-1} = \begin{bmatrix} \hat{\alpha}_{k|k-1} \\ \hat{q}_{k|k-1} \end{bmatrix} = E(x_k | y_{1:k-1}) = \begin{bmatrix} E(\alpha_k | y_{1:k-1}) \\ E(q_k | y_{1:k-1}) \end{bmatrix} \quad (9)$$

where $\hat{\alpha}_{k|k-1}$ and $\hat{q}_{k|k-1}$ are the one step forward prediction of the two state variables, respectively.

$$S_{k|k-1} = cov(x_k | y_{1:k-1}) = \begin{bmatrix} var(\alpha_k | y_{1:k-1}) & cov(\alpha_k, q_k | y_{1:k-1}) \\ cov(\alpha_k, q_k | y_{1:k-1}) & var(q_k | y_{1:k-1}) \end{bmatrix} \quad (10)$$

where $cov(\alpha_k, q_k | y_{1:k-1})$ is the predicted covariance of the two state variables, $var(\alpha_k | y_{1:k-1})$ and $var(q_k | y_{1:k-1})$ are the predicted variance of the two state variables, respectively. Based on Eqs. (5)–(10), when the battery capacity data at the k th cycle are obtained, the Kalman filter approach can be applied to estimate the state space variables, which includes the following five steps:

(1) State estimation:

$$\hat{x}_{k|k-1} = A_k \hat{x}_{k-1|k-1} \quad (11)$$

(2) Covariance estimation:

$$S_{k|k-1} = A_k S_{k-1|k-1} A_k^T + \begin{bmatrix} \sigma_\varphi^2 & 0 \\ 0 & \beta^2 (t_k - t_{k-1}) \end{bmatrix} \quad (12)$$

(3) Calculate the Kalman filter gain K_k :

$$K_k = S_{k|k-1} C_k^T (C_k S_{k|k-1} C_k^T + \sigma_\nu^2)^{-1} \quad (13)$$

(4) State update:

$$\hat{x}_{k|k} = \hat{x}_{k-1|k-1} + K_k (y_k - C_k \hat{x}_{k|k-1}) \quad (14)$$

(5) Covariance update:

$$S_{k|k} = S_{k|k-1} - K_k C_k S_{k|k-1} \quad (15)$$

The state space variables can be estimated and updated according to Eqs. (11)–(15), when the initial state $\hat{x}_{0|0}$ and the initial covariance $S_{0|0}$ of the battery degradation process are determined. It can be known that the parameters in the degradation model are dynamically estimated only by battery capacity data, which does not need the corresponding RUL labels.

3.2. Adaptive parameter estimation based on EM algorithm

Aiming to recognize unknown parameters in the adaptive state update model based on KF, this paper applies the expectation maximization (EM) algorithm for adaptive parameter estimation with the aid of the battery capacity data from the initial state to the current moment. The unknown parameter vector can be expressed as: $\Theta = [\hat{x}_{0|0}, S_{0|0}, \sigma_\varphi^2, \beta^2, \sigma_\nu^2]$. It should be noted that the essence of the EM algorithm is to approximate the maximum likelihood parameters estimation by maximizing the joint likelihood function under the premise of the posterior probability [39]. Based on the battery capacity data from the initial state to the k th working cycle $y_{1:k} = [y_1, y_2, \dots, y_k]$, the joint log-likelihood function $L_k(\Theta)$ can be constructed:

$$\begin{aligned} L_k(\Theta) &= \ln p(x_{0:k}, y_{1:k} | \Theta) \\ &= \ln [p(x_{0:k} | \Theta) p(y_{1:k} | x_{0:k}, \Theta)] \\ &= \ln p(x_{0:k} | \Theta) + \ln p(y_{1:k} | x_{0:k}, \Theta) \\ &= \ln p(x_0 | \Theta) + \ln \prod_{i=1}^k p(x_i | x_{i-1}, \Theta) + \ln \prod_{i=1}^k p(y_i | x_i, \Theta) \\ &= \ln p(x_0 | \Theta) + \sum_{i=1}^k \ln p(x_i | x_{i-1}, \Theta) + \sum_{i=1}^k \ln p(y_i | x_i, \Theta) \end{aligned} \quad (16)$$

Based on the KF adaptive state update model in Section 3.1, the probability distributions in Eq. (16) can be expressed as Eqs. (17)–(19):

$$x_0 | \Theta \sim N(\hat{x}_{0|0}, S_{0|0}) \quad (17)$$

$$x_i | x_{i-1}, \Theta \sim N(A_i x_{i-1}, Q_i) \quad (18)$$

$$y_i | x_i, \Theta \sim N(C_i x_i, \sigma_\nu^2) \quad (19)$$

Ignoring the constant term, Eqs. (17)–(19) are substituted into Eq. (16), and the joint log-likelihood function can be further derived:

$$\begin{aligned} L_k(\theta) = & \ln p(x_0|\theta) + \sum_{i=1}^k \ln p(x_i|x_{i-1}, \theta) + \sum_{i=1}^k \ln p(y_i|x_i, \theta) \\ & \propto -\frac{1}{2} \left[\ln |S_{0|0}| + (x_0 - \hat{x}_{0|0})^T (S_{0|0})^{-1} (x_0 - \hat{x}_{0|0}) \right] \\ & - \frac{1}{2} \sum_{i=1}^k \left[\ln |Q_i| + (x_i - Ax_{i-1})^T (Q_i)^{-1} (x_i - Ax_{i-1}) \right] \\ & - \frac{1}{2} \sum_{i=1}^k \left[\ln \sigma_v^2 + \frac{(y_i - C_i x_i)^2}{\sigma_v^2} \right] \end{aligned} \quad (20)$$

For the j th iteration process in EM algorithm, the unknown parameter vector can be expressed as: $\hat{\theta}_k^{(j)} = [\hat{x}_{0|0k}^{(j)}, S_{0|0k}^{(j)}, (\sigma_\varphi^2)^{(j)}, (\beta^2)^{(j)}, (\sigma_v^2)^{(j)}]$. According to the principle of the EM algorithm, the $j+1$ th iteration process consists of E step and M step:

(1) E step:

$$L_k(\theta|\hat{\theta}_k^{(j)}) = E_{x_{0:k}|y_{1:k}, \hat{\theta}_k^{(j)}} [L_k(\theta)] \quad (21)$$

(2) M step:

$$\hat{\theta}_k^{(j+1)} = \underset{\theta}{\operatorname{argmax}} \left\{ E_{x_{0:k}|y_{1:k}, \hat{\theta}_k^{(j)}} [L_k(\theta)] \right\} \quad (22)$$

E step and M step are iteratively performed until the convergence criterion is met, thereby obtaining the unknown parameter of the adaptive state model. According to Eq. (20), Eq. (21) can be extended to the Eq. (23). In Eq. (23), $E_{x_{0:k}|y_{1:k}, \hat{\theta}_k^{(j)}}(x_i)$, $E_{x_{0:k}|y_{1:k}, \hat{\theta}_k^{(j)}}(x_i x_i^T)$,

$E_{x_{0:k}|y_{1:k}, \hat{\theta}_k^{(j)}}(x_i x_{i-1}^T)$ and $E_{x_{0:k}|y_{1:k}, \hat{\theta}_k^{(j)}}(x_{i-1} x_i^T)$ are the conditional expectations of the state space variables based on the battery capacity data $y_{1:k} = [y_1, y_2, \dots, y_k]$, which are calculated with the Rauch–Tung–Striebel (RTS) smoothing algorithm. For linear state update model with Gaussian measurement noise, the RTS algorithm is the optimal state estimator with minimum mean square error, which has wide applications in signal processing, aerospace, and target tracking [40,41].

Based on the forward iteration of the KF, the expectation of the state variables $\tilde{x}_{k|k}$ and covariance $\tilde{S}_{k|k}$ can be estimated:

$$\tilde{x}_{k|k} = \hat{x}_{k|k} \quad (24)$$

$$\tilde{S}_{k|k} = S_{k|k} \quad (25)$$

Correspondingly, the covariance between the $k-1$ th and the k th state variables can be calculated:

$$\tilde{S}_{k-1,k|k} = \operatorname{cov}(x_{k-1}, x_k | y_{1:k}) = A_k S_{k-1|k-1} - K_k C_k A_k S_{k-1|k-1} \quad (26)$$

$$\begin{aligned} L_k(\theta|\hat{\theta}_k^{(j)}) = & E_{x_{0:k}|y_{1:k}, \hat{\theta}_k^{(j)}} [L_k(\theta)] \\ & \propto E_{x_{0:k}|y_{1:k}, \hat{\theta}_k^{(j)}} \left\{ -\frac{1}{2} \left[\ln |S_{0|0}| + (x_0 - \hat{x}_{0|0})^T (S_{0|0})^{-1} (x_0 - \hat{x}_{0|0}) \right] - \frac{1}{2} \sum_{i=1}^k \left[\ln |Q_i| + (x_i - Ax_{i-1})^T (Q_i)^{-1} (x_i - Ax_{i-1}) \right] \right. \\ & \left. - \frac{1}{2} \sum_{i=1}^k \left[\ln \sigma_v^2 + \frac{(y_i - C_i x_i)^2}{\sigma_v^2} \right] \right\} \\ & \propto -\frac{1}{2} \ln |S_{0|0k}^{(j)}| - \frac{k}{2} \ln |Q_k^{(j)}| - \frac{k}{2} \ln (\sigma_\varphi^2)^{(j)} - \frac{1}{2} \operatorname{tr} \left\{ (S_{0|0k}^{(j)})^{-1} E_{x_{0:k}|y_{1:k}, \hat{\theta}_k^{(j)}} \left[x_0 x_0^T - x_0 \left(\hat{x}_{0|0k}^{(j)} \right)^T - \hat{x}_{0|0k}^{(j)} x_0 + \hat{x}_{0|0k}^{(j)} \left(\hat{x}_{0|0k}^{(j)} \right)^T \right] \right\} \\ & - \frac{1}{2} \sum_{i=1}^k \operatorname{tr} \left\{ (Q_i^{(j)})^{-1} E_{x_{0:k}|y_{1:k}, \hat{\theta}_k^{(j)}} \left[x_i x_i^T - x_i x_{i-1}^T A_i^T - A_i x_{i-1} x_i^T + (A_i x_{i-1}) (A_i x_{i-1})^T \right] \right\} \\ & - \frac{1}{2} \sum_{i=1}^k \frac{1}{(\sigma_v^2)^{(j)}} E_{x_{0:k}|y_{1:k}, \hat{\theta}_k^{(j)}} [y_i^2 - 2y_i C_i x_i + C_i x_i x_i^T C_i^T] \end{aligned} \quad (23)$$

According to the initialization conditions in Eqs. (24)–(26), RTS smoothing can be performed. The gain of RTS smoothing is shown in Eq. (27):

$$G_i = S_{i|i} A_i^T (S_{i+1|i})^{-1} \quad (27)$$

Correspondingly, the expectation of $\tilde{x}_{i|k}$ and $\tilde{S}_{i|k}$ in the backward iteration process can be calculated by:

$$\tilde{x}_{i|k} = \hat{x}_{i|i} + G_i (\tilde{x}_{i+1|k} - \hat{x}_{i+1|i}) \quad (28)$$

$$\tilde{S}_{i|k} = S_{i|i} + G_i (\tilde{S}_{i+1|k} - S_{i+1|i}) G_i^T \quad (29)$$

In addition, the covariance between two adjacent cyclic state matrices can be calculated by:

$$\tilde{S}_{i-1,i|k} = S_{i|i} G_{i-1}^T + G_i (\tilde{S}_{i,i+1|k} - A_i S_{i|i}) G_{i-1}^T \quad (30)$$

Based on Eqs. (24)–(30), the conditional expectations can be calculated by:

$$E_{x_{0:k}|y_{1:k}, \hat{\theta}_k^{(j)}}(x_i) = \tilde{x}_{i|k} \quad (31)$$

$$E_{x_{0:k}|y_{1:k}, \hat{\theta}_k^{(j)}}(x_i x_i^T) = \tilde{x}_{i|k} \tilde{x}_{i|k}^T + \tilde{S}_{i|k} \quad (32)$$

$$E_{x_{0:k}|y_{1:k}, \hat{\theta}_k^{(j)}}(x_i x_{i-1}^T) = \tilde{x}_{i|k} \tilde{x}_{i-1|k}^T + \tilde{S}_{i-1,i|k} \quad (33)$$

$$E_{x_{0:k}|y_{1:k}, \hat{\theta}_k^{(j)}}(x_{i-1} x_i^T) = \tilde{x}_{i-1|k} \tilde{x}_{i|k}^T + \tilde{S}_{i-1,i|k} \quad (34)$$

The complete RTS smoothing algorithm is summarized in Algorithm 1.

Algorithm 1 RTS smoothing algorithm

Input: The battery capacity data $y_{1:k} = [y_1, y_2, \dots, y_k]$

Output: The conditional expectations in Eqs. (31)–(34)

- 1: Calculate $\tilde{x}_{k|k}$, $\tilde{S}_{k|k}$ and $\tilde{S}_{k-1,k|k}$ according to Eqs. (24)–(26).
- 2: Calculate the RTS smoothing gain G_i by Eq. (27).
- 3: Update the state and covariance to obtain $\tilde{x}_{i|k}$, $\tilde{S}_{i|k}$ and $\tilde{S}_{i-1,i|k}$ according to Eqs. (28)–(30).
- 4: Calculate the conditional expectations by Eqs. (31)–(34).

Substituting the conditional expectations in Eqs. (31)–(34) into Eq. (23), Eq. (35) can be derived.

On the basis of E step in Eqs. (23)–(35), M step of EM algorithm can be calculated according to Eq. (22). For the unknown parameters of the degradation model, the first-order partial derivative of each parameter in vector $\hat{\theta}_k^{(j)} = [\hat{x}_{0|0k}^{(j)}, S_{0|0k}^{(j)}, (\sigma_\varphi^2)^{(j)}, (\beta^2)^{(j)}, (\sigma_v^2)^{(j)}]$ is set to 0. In this sense, the results of Eqs. (36)–(39) can be obtained:

$$\hat{x}_{0|0k}^{(j+1)} = \tilde{x}_{0|k} \quad (36)$$

$$\begin{aligned}
L_k(\theta | \hat{\theta}_k^{(j)}) &= E_{x_{0:k}|y_{1:k}, \hat{\theta}_k^{(j)}} [L_k(\theta)] \\
&\propto -\frac{1}{2} \ln |S_{0|0k}^{(j)}| - \frac{k}{2} \ln |Q_k^{(j)}| - \frac{k}{2} \ln (\sigma_v^2)^{(j)} - \frac{1}{2} \text{tr} \left\{ \left(S_{0|0k}^{(j)} \right)^{-1} \left[\left(\tilde{x}_{0|k} \tilde{x}_{0|k}^T + \tilde{S}_{0|k} \right) - \left(\tilde{x}_{0|k} \right) \left(\hat{x}_{0|0k}^{(j)} \right)^T - \left(\hat{x}_{0|0k}^{(j)} \right) \left(\tilde{x}_{0|k} \right)^T + \left(\hat{x}_{0|0k}^{(j)} \right) \left(\hat{x}_{0|0k}^{(j)} \right)^T \right] \right\} \\
&\quad - \frac{1}{2} \sum_{i=1}^k \text{tr} \left\{ \left(Q_k^{(j)} \right)^{-1} \left[\left(\tilde{x}_{i|k} \tilde{x}_{i|k}^T + \tilde{S}_{i|k} \right) - \left(\tilde{x}_{i|k} \tilde{x}_{i-1|k}^T + \tilde{S}_{i-1, i|k} \right) A_i^T - A_i \left(\tilde{x}_{i-1|k} \tilde{x}_{i|k}^T + \tilde{S}_{i-1, i|k} \right) + A_i \left(\tilde{x}_{i-1|k} \tilde{x}_{i-1|k}^T + \tilde{S}_{i-1|k} \right) A_i^T \right] \right\} \\
&\quad - \frac{1}{2 (\sigma_v^2)^{(j)}} \sum_{i=1}^k \left[y_i^2 - 2 y_i C_i \tilde{x}_{i|k} + C_i \left(\tilde{x}_{i|k} \tilde{x}_{i|k}^T + \tilde{S}_{i|k} \right) C_i^T \right]
\end{aligned} \tag{35}$$

$$S_{0|0k}^{(j+1)} = \tilde{S}_{0|k} \tag{37}$$

$$\begin{aligned}
Q_k^{(j+1)} &= \frac{1}{k} \sum_{i=1}^k \left[\left(\tilde{x}_{i|k} \tilde{x}_{i|k}^T + \tilde{S}_{i|k} \right) - \left(\tilde{x}_{i|k} \tilde{x}_{i-1|k}^T + \tilde{S}_{i-1, i|k} \right) A_i^T \right. \\
&\quad \left. - A_i \left(\tilde{x}_{i-1|k} \tilde{x}_{i|k}^T + \tilde{S}_{i-1, i|k} \right) + A_i \left(\tilde{x}_{i-1|k} \tilde{x}_{i-1|k}^T + \tilde{S}_{i-1|k} \right) A_i^T \right]
\end{aligned} \tag{38}$$

$$(\sigma_v^2)^{(j+1)} = \frac{1}{k} \sum_{i=1}^k \left[y_i^2 - 2 y_i C_i \tilde{x}_{i|k} + C_i \left(\tilde{x}_{i|k} \tilde{x}_{i|k}^T + \tilde{S}_{i|k} \right) C_i^T \right] \tag{39}$$

where $Q_k^{(j+1)} = \begin{bmatrix} (\sigma_\varphi^2)^{(j+1)} & 0 \\ 0 & (\beta^2)^{(j+1)} (t_k - t_{k-1}) \end{bmatrix}$. The iteration of EM algorithm is stopped until $\|\hat{\theta}_k^{(j+1)} - \hat{\theta}_k^{(j)}\|_1 < \epsilon$ is satisfied or the total number of iterations reaches maximum, where ϵ is the threshold with a small value. In this sense, the adaptive estimation of unknown parameters is achieved in the degradation model. For each working cycle of the battery, the unknown parameters of the adaptive estimation model are updated based on all the battery capacity data $y_{1:k}$, which overcomes the constraints of the strong Markov characteristics. The whole flow of the parameter estimation based on EM algorithm can be summarized in Algorithm 2.

Algorithm 2 Parameter estimation based on EM algorithm

Input: The battery capacity data $y_{1:k} = [y_1, y_2, \dots, y_k]$

Output: The parameter estimation results in Eqs. (36)–(39)

- 1: Initialize the total number of iterations J , threshold ϵ , the serial number of iteration $j = 1$.
- 2: **while** $j \leq J$ or $\|\hat{\theta}_k^{(j+1)} - \hat{\theta}_k^{(j)}\|_1 < \epsilon$ **do**
- 3: Construct the joint log-likelihood function under the posterior probability according to Eq. (23) (E step).
- 4: Obtain the conditional expectations by RTS smoothing algorithm in Eqs. (24)–(34) (E step).
- 5: Substitute the conditional expectations into Eq. (23) to obtain Eq. (35) (E step).
- 6: Set the first-order partial derivative of each parameter in unknown parameter vector to zero, and obtain the update results in Eqs. (36)–(39) (M step).
- 7: $j \rightarrow j + 1$.
- 8: **end while**

3.3. Convergence analysis of adaptive parameter estimation

This subsection further discusses the convergence of EM adaptive parameter estimation in the iterative process and whether the estimation results can converge to the globally optimal solution.

Lemma 1 ([42]). Let the unknown parameter vector obtained by the j th iteration of the EM algorithm be expressed as $\hat{\theta}_k^{(j+1)} = [\hat{x}_{0|0k}^{(j+1)}, S_{0|0k}^{(j+1)}, (\sigma_\varphi^2)^{(j+1)}, (\beta^2)^{(j+1)}, (\sigma_v^2)^{(j+1)}]$. Then, the inequality $L_k(\theta | \hat{\theta}_k^{(j+1)}) \geq L_k(\theta | \hat{\theta}_k^{(j)})$ always holds, and the equality sign is reached only when $\hat{\theta}_k^{(j+1)} = \hat{\theta}_k^{(j)}$.

Proof. Please refer to [42] Section 4 for the proof.

In Section 3.2, according to Eq. (22), the purpose of the iteration of Eqs. (36)–(39) is to maximize $L_k(\theta | \hat{\theta}_k^{(j)})$ in Eq. (35). If the estimation result can converge to the globally optimal solution in the iterative process, it is necessary to prove that $\hat{\theta}_k^{(j+1)}$ in the j th iteration is equal to the value that makes $L_k(\theta | \hat{\theta}_k^{(j)})$ reach the maximum. In other words, it needs to prove that the second-order partial derivative matrix in Eq. (35) is negative definite, when $\theta = \hat{\theta}_k^{(j+1)}$. For this purpose, the derivative matrix is written as Eq. (40) which is given in Box I, where $P_1 \sim P_6$ are shown in Eqs. (41)–(46), respectively:

$$P_1 = \text{tr} \left\{ \left(\tilde{S}_{0|k} \right)^{-1} \left[\left(\tilde{S}_{0|k} \right)^{-1} \right]^T \left(\hat{x}_{0|0k}^{(j+1)} - \tilde{x}_{0|k} \right) \right\} \tag{41}$$

$$P_2 = \left(\tilde{S}_{0|k} \right)^{-1} \left[\left(\tilde{S}_{0|k} \right)^{-1} \right]^T \tag{42}$$

$$\begin{aligned}
P_3 &= \left(\tilde{x}_{0|k} \tilde{x}_{0|k}^T + \tilde{S}_{0|k} \right) - \left(\tilde{x}_{0|k} \right) \left(\hat{x}_{0|0k}^{(j+1)} \right)^T - \left(\hat{x}_{0|0k}^{(j+1)} \right) \left(\tilde{x}_{0|k} \right)^T \\
&\quad + \left(\hat{x}_{0|0k}^{(j+1)} \right) \left(\hat{x}_{0|0k}^{(j+1)} \right)^T
\end{aligned} \tag{43}$$

$$P_4 = \left(Q_k^{(j+1)} \right)^{-1} \left[\left(Q_k^{(j+1)} \right)^{-1} \right]^T \tag{44}$$

$$\begin{aligned}
P_5 &= \sum_{i=1}^k \left[\left(\tilde{x}_{i|k} \tilde{x}_{i|k}^T + \tilde{S}_{i|k} \right) - \left(\tilde{x}_{i|k} \tilde{x}_{i-1|k}^T + \tilde{S}_{i-1, i|k} \right) A_i^T \right. \\
&\quad \left. - A_i \left(\tilde{x}_{i-1|k} \tilde{x}_{i|k}^T + \tilde{S}_{i-1, i|k} \right) + A_i \left(\tilde{x}_{i-1|k} \tilde{x}_{i-1|k}^T + \tilde{S}_{i-1|k} \right) A_i^T \right]
\end{aligned} \tag{45}$$

$$P_6 = \sum_{i=1}^k \left[y_i^2 - 2 y_i C_i \tilde{x}_{i|k} + C_i \left(\tilde{x}_{i|k} \tilde{x}_{i|k}^T + \tilde{S}_{i|k} \right) C_i^T \right] \tag{46}$$

Based on Eqs. (40)–(46), for the j th iteration process, combined with Eqs. (36)–(39), the leading principal minors of Eq. (40) can be calculated as Eqs. (47)–(50):

$$A_1|_{\theta=\hat{\theta}_k^{(j+1)}} = -\text{tr} \left[\left(\tilde{S}_{0|k} \right)^{-1} \right] < 0 \tag{47}$$

$$\begin{aligned}
A_2|_{\theta=\hat{\theta}_k^{(j+1)}} &= -\text{tr} \left[\left(\tilde{S}_{0|k} \right)^{-1} \right] \left\{ \frac{1}{2} \text{tr} (P_2) - \text{tr} \left[P_2 \left(\tilde{S}_{0|k} \right)^{-1} P_3 \right] \right\} - P_1^2 \\
&= -\text{tr} \left[\left(\tilde{S}_{0|k} \right)^{-1} \right] \left[-\frac{1}{2} \text{tr} (P_2) \right] > 0
\end{aligned} \tag{48}$$

$$\begin{aligned}
A_3|_{\theta=\hat{\theta}_k^{(j+1)}} &= \left\{ \frac{k}{2} \text{tr} (P_4) - \text{tr} \left[P_4 \left(Q_k^{(j+1)} \right)^{-1} P_5 \right] \right\} A_2 \Big|_{\theta=\hat{\theta}_k^{(j+1)}} \\
&= -\frac{k^3}{2} \text{tr} \left\{ \left(P_5 \right)^{-1} \left[\left(P_5 \right)^{-1} \right]^T \right\} A_2 \Big|_{\theta=\hat{\theta}_k^{(j+1)}} < 0
\end{aligned} \tag{49}$$

$$\begin{aligned}
\frac{\partial^2 L_k(\theta | \hat{\theta}_k^{(j)})}{\partial \theta \partial \theta^T} \Big|_{\theta = \hat{\theta}_k^{(j+1)}} &= \begin{bmatrix} \frac{\partial^2 L_k(\theta | \hat{\theta}_k^{(j)})}{\partial^2 \tilde{x}_{0|0k}^{(j)}} \Big|_{\theta = \hat{\theta}_k^{(j+1)}} & \frac{\partial^2 L_k(\theta | \hat{\theta}_k^{(j)})}{\partial \tilde{x}_{0|0k}^{(j)} \partial S_{0|0k}^{(j)}} \Big|_{\theta = \hat{\theta}_k^{(j+1)}} & \frac{\partial^2 L_k(\theta | \hat{\theta}_k^{(j)})}{\partial \tilde{x}_{0|0k}^{(j)} \partial Q_k^{(j)}} \Big|_{\theta = \hat{\theta}_k^{(j+1)}} & \frac{\partial^2 L_k(\theta | \hat{\theta}_k^{(j)})}{\partial \tilde{x}_{0|0k}^{(j)} \partial (\sigma_v^{(j)})_k} \Big|_{\theta = \hat{\theta}_k^{(j+1)}} \\ \frac{\partial^2 L_k(\theta | \hat{\theta}_k^{(j)})}{\partial S_{0|0k}^{(j)} \partial \tilde{x}_{0|0k}^{(j)}} \Big|_{\theta = \hat{\theta}_k^{(j+1)}} & \frac{\partial^2 L_k(\theta | \hat{\theta}_k^{(j)})}{\partial^2 S_{0|0k}^{(j)}} \Big|_{\theta = \hat{\theta}_k^{(j+1)}} & \frac{\partial^2 L_k(\theta | \hat{\theta}_k^{(j)})}{\partial S_{0|0k}^{(j)} \partial Q_k^{(j)}} \Big|_{\theta = \hat{\theta}_k^{(j+1)}} & \frac{\partial^2 L_k(\theta | \hat{\theta}_k^{(j)})}{\partial S_{0|0k}^{(j)} \partial (\sigma_v^{(j)})_k} \Big|_{\theta = \hat{\theta}_k^{(j+1)}} \\ \frac{\partial^2 L_k(\theta | \hat{\theta}_k^{(j)})}{\partial Q_k^{(j)} \partial \tilde{x}_{0|0k}^{(j)}} \Big|_{\theta = \hat{\theta}_k^{(j+1)}} & \frac{\partial^2 L_k(\theta | \hat{\theta}_k^{(j)})}{\partial Q_k^{(j)} \partial S_{0|0k}^{(j)}} \Big|_{\theta = \hat{\theta}_k^{(j+1)}} & \frac{\partial^2 L_k(\theta | \hat{\theta}_k^{(j)})}{\partial^2 Q_k^{(j)}} \Big|_{\theta = \hat{\theta}_k^{(j+1)}} & \frac{\partial^2 L_k(\theta | \hat{\theta}_k^{(j)})}{\partial Q_k^{(j)} \partial (\sigma_v^{(j)})_k} \Big|_{\theta = \hat{\theta}_k^{(j+1)}} \\ \frac{\partial^2 L_k(\theta | \hat{\theta}_k^{(j)})}{\partial (\sigma_v^{(j)})_k \partial \tilde{x}_{0|0k}^{(j)}} \Big|_{\theta = \hat{\theta}_k^{(j+1)}} & \frac{\partial^2 L_k(\theta | \hat{\theta}_k^{(j)})}{\partial (\sigma_v^{(j)})_k \partial S_{0|0k}^{(j)}} \Big|_{\theta = \hat{\theta}_k^{(j+1)}} & \frac{\partial^2 L_k(\theta | \hat{\theta}_k^{(j)})}{\partial (\sigma_v^{(j)})_k \partial Q_k^{(j)}} \Big|_{\theta = \hat{\theta}_k^{(j+1)}} & \frac{\partial^2 L_k(\theta | \hat{\theta}_k^{(j)})}{\partial^2 (\sigma_v^{(j)})_k} \Big|_{\theta = \hat{\theta}_k^{(j+1)}} \end{bmatrix} \\
&= \begin{bmatrix} -tr[(\tilde{S}_{0|0k})^{-1}] & P_1 & 0 & 0 \\ P_1 & \frac{1}{2} tr(P_2) - tr[P_2(\tilde{S}_{0|0k})^{-1}P_3] & 0 & 0 \\ 0 & 0 & \frac{k}{2} tr(P_4) - tr[P_4(Q_k^{(j+1)})^{-1}P_5] & 0 \\ 0 & 0 & 0 & \frac{k}{2} [(\sigma_v^{(j+1)})_k]^{-2} - [(\sigma_v^{(j+1)})_k]^{-3} P_6 \end{bmatrix}
\end{aligned} \quad (40)$$

Box 1.

$$\begin{aligned}
A_4|_{\theta = \hat{\theta}_k^{(j+1)}} &= \left\{ \frac{k}{2} [(\sigma_v^{(j+1)})_k]^{-2} - [(\sigma_v^{(j+1)})_k]^{-3} P_6 \right\} A_3 \Big|_{\theta = \hat{\theta}_k^{(j+1)}} \\
&= \frac{k(\sigma_v^{(j+1)})_k - 2P_6}{2[(\sigma_v^{(j+1)})_k]^3} A_3 \Big|_{\theta = \hat{\theta}_k^{(j+1)}} \\
&= -\frac{k^3}{2P_6^2} A_3 \Big|_{\theta = \hat{\theta}_k^{(j+1)}} > 0
\end{aligned} \quad (50)$$

It can be learned that $\hat{\theta}_k^{(j+1)}$ is the only value that satisfies $\frac{\partial L_k(\theta | \hat{\theta}_k^{(j)})}{\partial \theta} = 0$. Furthermore, according to Eqs. (47)–(50), the matrix in Eq. (40) is negative definite when $\theta = \hat{\theta}_k^{(j+1)}$. Therefore, $\hat{\theta}_k^{(j+1)}$ is the maximum point of $L_k(\theta | \hat{\theta}_k^{(j)})$, which verifies that $\hat{\theta}_k^{(j+1)}$ is the globally optimal solution in the adaptive parameter estimation. Through the analysis of the convergence of the adaptive algorithm, it can be concluded that the unique maximum analytical solution can be obtained after the M step, which means that the EM-RTS algorithm proposed in this paper has good convergence.

3.4. Adaptive RUL prediction of battery

The essence of the battery's RUL is the first arrival time for the failure. In detail, it is defined as the time interval between the current state and the moment when battery's SOC drops to 80% for the first time, the rationale of which can be referred to the articles [32,37,38]. For the k th working cycle of the battery, the RUL can be written as Eq. (51):

$$RUL_k = \inf \{l_k : q(t_k + l_k) \leq 0.8q_0 \mid q_0 > 0\} \quad (51)$$

where RUL_k is the infimum of the l_k , t_k is the k th working cycle, and q_0 is the capacity of the battery in the initial state.

Lemma 2 ([43]). For the linear Winner process $\{q(t), t \geq 0\}$ in Eq. (1), the probability density function $f_T(t)$ of the first time that $q(t)$ reaches the failure threshold is shown in Eq. (52):

$$f_T(t) = \frac{0.8q_0}{\sqrt{2\pi t^3 \beta^2}} \exp \left\{ -\frac{(0.8q_0 - \alpha t)^2}{2\beta^2 t} \right\} \quad (52)$$

Proof. Please refer to [43] Section 1 for the proof.

In this paper, it is assumed that the drift coefficient α obeys normal distribution, the probability density distribution of α_k estimated from the battery capacity data $y_{1:k} = [y_1, y_2, \dots, y_k]$ is:

$$p(\alpha_k \mid y_{1:k}) = \frac{1}{\sqrt{2\pi var(\alpha_k \mid y_{1:k})}} \exp \left\{ -\frac{(\alpha_k - E(\alpha_k \mid y_{1:k}))^2}{2var(\alpha_k \mid y_{1:k})} \right\} \quad (53)$$

where $E(\alpha_k \mid y_{1:k})$ and $var(\alpha_k \mid y_{1:k})$ are the posterior expectation and variance of α_k in the dynamic state space model, respectively. Combining Eqs. (52)–(53), the probability density function (PDF) of the RUL l_k can be calculated as:

$$\begin{aligned}
f_{RUL_k|y_{1:k}}(l_k \mid y_{1:k}) &= \int_0^\infty f_{RUL_k|\alpha_k, y_{1:k}}(l_k \mid \alpha_k, y_{1:k}) p(\alpha_k \mid y_{1:k}) d\alpha_k \\
&= \frac{y_k - 0.8q_0}{\sqrt{2\pi l_k^3 (S_{k|k}^{(\alpha)} l_k + \beta_k^2)}} \exp \left\{ -\frac{(y_k - 0.8q_0 - x_{k|k}^{(\alpha)} l_k)^2}{2l_k (S_{k|k}^{(\alpha)} l_k + \beta_k^2)} \right\}
\end{aligned} \quad (54)$$

where $x_{k|k}^{(\alpha)}$ and $S_{k|k}^{(\alpha)}$ are the posterior expectation and variance of α_k under the dynamic state space model, respectively, β_k is the variation coefficient acquired by EM algorithm. It can be known that the distribution of model parameter α_k is integrated in the expression of probability density function, which describes the uncertainty of the parameter estimation. On this basis, the battery's RUL is the mathematical expectation for Eq. (54), which can be shown in Eq. (55):

$$RUL_k = \int_0^\infty \int_0^\infty f_{RUL_k|\alpha_k, y_{1:k}}(l_k \mid \alpha_k, y_{1:k}) p(\alpha_k \mid y_{1:k}) l_k dl_k d\alpha_k \quad (55)$$

The adaptive prediction approach proposed in this paper can be summarized in Algorithm 3. To better represent the adaptive RUL prediction approach, the flow of which is shown in Fig. 1. It should be pointed out that the dotted line in Fig. 1 means the estimation result of the unknown parameter at the k th working cycle is fed back to the state estimation and covariance update under the $k+1$ th working cycle.

4. Case study

4.1. Data description and evaluation indicators

The battery dataset provided by NASA Ames research center is used to verify the adaptive RUL prediction approach. The whole dataset

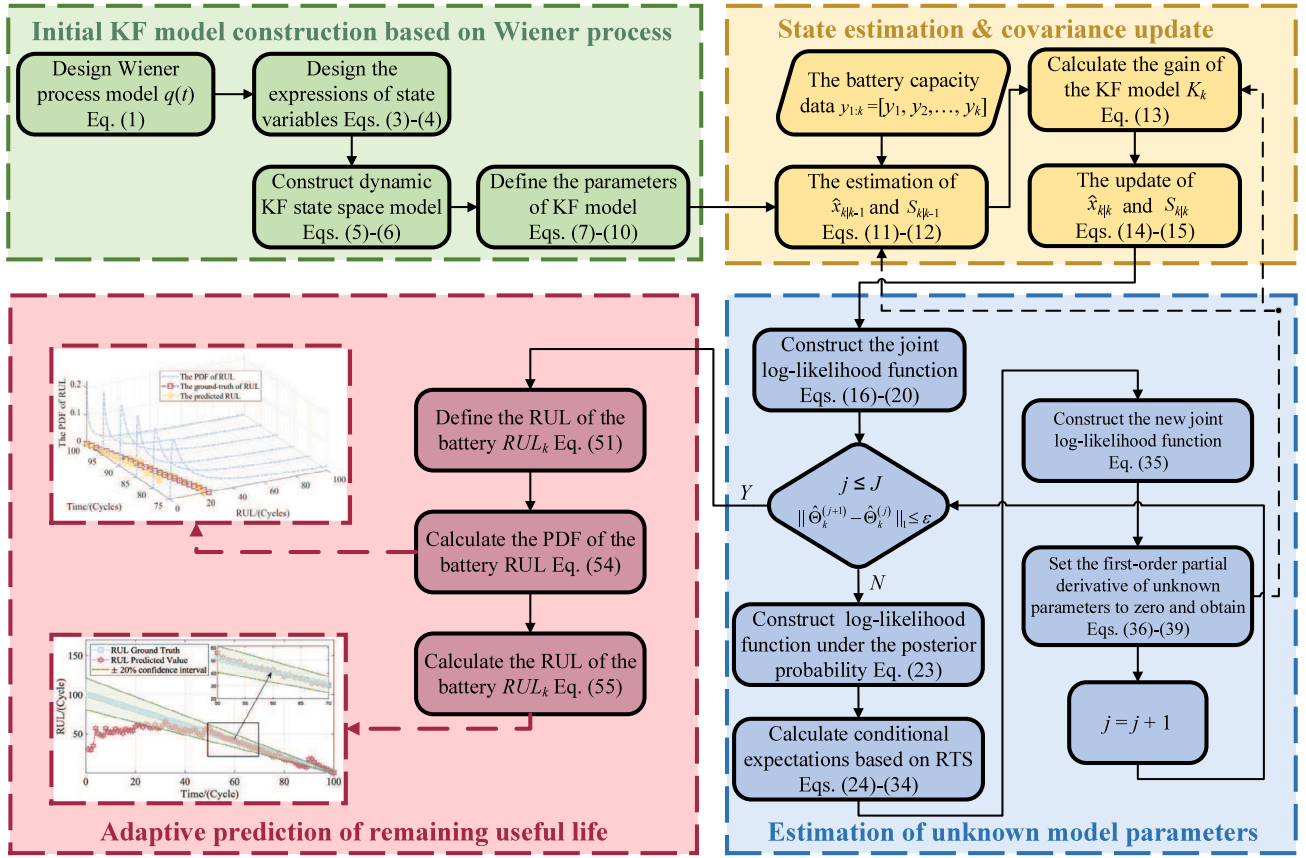


Fig. 1. The flow of RUL prediction approach.

Algorithm 3 Adaptive RUL prediction algorithm**Input:** The battery capacity data $y_{1:k} = [y_1, y_2, \dots, y_k]$ **Output:** The RUL_k under the k th working cycle and the corresponding probability density function $f_{RUL_k|y_{1:k}}(l_k | y_{1:k})$

- 1: Construct adaptive KF state update model of the battery according to Eqs. (1)–(6).
- 2: Define state and covariance matrices of KF state update model according to Eqs. (7)–(10).
- 3: Update KF model to obtain $\hat{x}_{k|k}$ and $S_{k|k}$ by Eqs. (11)–(15).
- 4: Construct joint log-likelihood function by Eqs. (16)–(20).
- 5: The parameter estimation of the unknown vector according to Algorithm 2.
- 6: Define the battery's RUL according to Eq. (51).
- 7: Obtain the probability density function of the RUL and the corresponding expectation according to Eqs. (54)–(55).

contains four 18650 lithium-ion batteries (numbered B0005, B0006, B0007, and B0018), the rated capacity of which are 2 Ah. The four batteries work in three modes: the charging mode, the discharging mode and the impedance test mode at 24 °C. For the charging mode, the battery is charged with a constant current of 1.5 A until the voltage reaches 4.2 V. In addition, the battery is continuously charged with a constant voltage until the charging current drops to 20 mA. For the discharging mode, the battery is discharged until the voltage of the four batteries drops to 2.7 V, 2.5 V, 2.2 V, and 2.5 V, respectively. The impedance test is to scan the electrochemical impedance spectroscopy (EIS) of the battery with frequency from 0.1 Hz to 5 kHz to acquire the internal parameters of the battery. The battery dataset directly provides the battery capacity at the end of each discharge cycle, which can be

used for adaptive prediction of the battery's RUL [44]. The degrading capacity curves of these batteries are shown in Fig. 2.

This paper uses root mean square error (RMSE), mean absolute error (MAE), and absolute error (AE) to evaluate the RUL prediction performance:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (RUL_{pi} - RUL_{ti})^2} \quad (56)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |RUL_{pi} - RUL_{ti}| \quad (57)$$

$$AE = |RUL_p - RUL_t| \quad (58)$$

where N is the number of working cycles, i is the index of the working cycles, RUL_{ti} and RUL_{pi} are the RUL ground truth and the predicted value of the i th cycle, respectively, RUL_t and RUL_p are the RUL ground truth and the predicted value of a specific cycle, respectively. The smaller value indicates the better RUL prediction performance of the approach.

4.2. Experimental results of adaptive parameter estimation

One of the advantages of the proposed adaptive RUL prediction approach is that the parameters in the degradation model are dynamically estimated only by battery capacity data. In this paper, EM algorithm is applied for adaptive parameter estimation. In detail, the unknown parameter vector can be written in: $\theta = [\hat{x}_{0|0}, S_{0|0}, \sigma_\phi^2, \beta^2, \sigma_v^2]$, where

$$\hat{x}_{0|0} = [-0.01, 2]^T, \quad S_{0|0} = \begin{bmatrix} 0.01 & 0 \\ 0 & 0.01 \end{bmatrix}, \quad \sigma_\phi^2 = \beta^2 = \sigma_v^2 = 0.01.$$

According to the steps in Algorithm 3, the model parameters are updated with the aid of the battery capacity data in each working cycle. Taking the B0005 battery as an example, the adaptive parameter

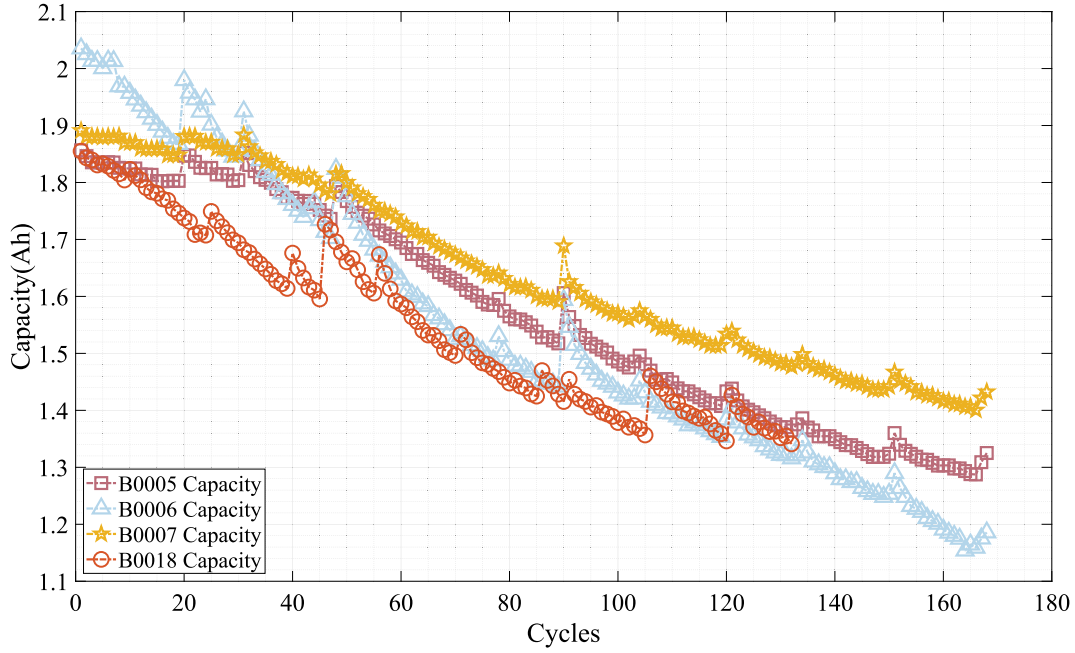


Fig. 2. The degrading capacity curves of four batteries.

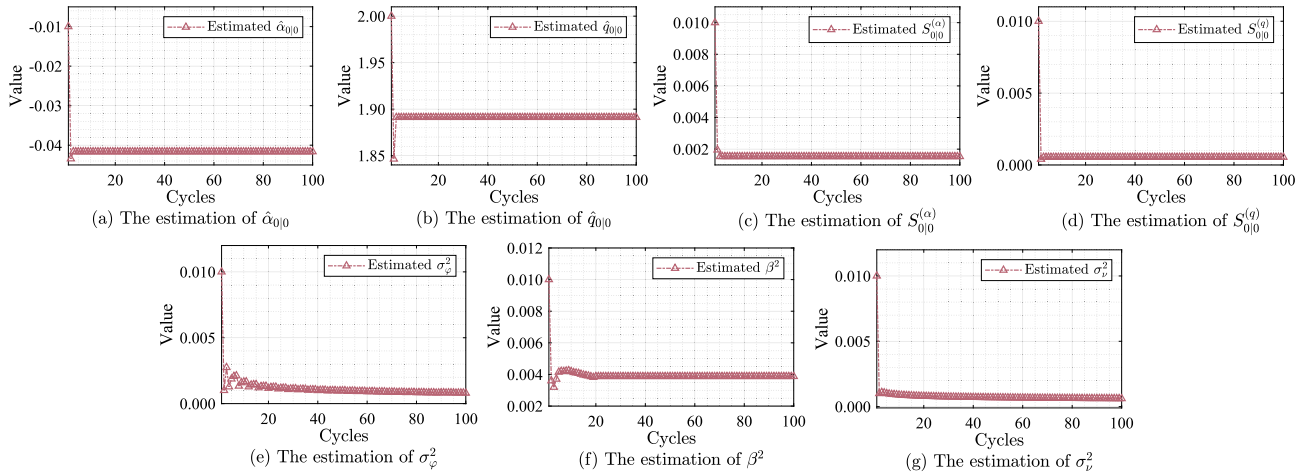


Fig. 3. The adaptive parameter estimation results of the B0005 battery.

estimation results are shown in Fig. 3. The figure represents that with the accumulation of battery capacity data, all the model parameters converge to a specific value within 10 working cycles. Therefore, the adaptive parameter estimation approach in this paper has an excellent convergence performance.

4.3. Experimental results of adaptive RUL prediction

As for the uncertainty of the specific failure time in the battery degradation process, the distribution of model parameter α_k is integrated in the expression of RUL probability density function in Eq. (54), in order to measure the uncertainty of the parameter estimation in the battery degradation model. The RUL probability density functions of the four batteries in the dataset under several different working cycles during the degradation process are shown in Fig. 4, along with the corresponding predicted RUL and ground-truth. It can be seen that with the accumulation of battery capacity data, the probability density function of the RUL becomes sharper, which indicates that the proposed RUL prediction approach has good adaptability. The reason is

that more and more battery capacity data are applied for the adaptive estimation of model parameters, which contributes to the reduction of the estimation uncertainty of the state space model parameter α_k .

Fig. 5 shows the experimental results of the predicted RUL and the RUL ground truth of each battery in the battery dataset under each working cycle. It should be noted that the green dashed line in the figure means the upper and lower bounds of the confidence interval for $\alpha = 0.2$. The green shaded area is the range of the corresponding confidence interval, which represents the tolerance of 20% predicted deviation. It can be learned that the predicted RUL can basically fall within the confidence interval of each battery in the later cycles of the degradation process.

To reflect the superiority of the proposed approach for predicting the RUL of the batteries, this paper compares the proposed approach with some existing approaches, namely, Logistic regression (LR), decision tree regression (DTR), extreme gradient boosting (XGBoost), random forest regression (RFR), adaptive boosting (AdaBoost), partial least squares regression (PLS), gradient boosting regression (GBR), support vector machine regression (SVR), K-nearest neighbor (KNN), light gradient boosting machine (LightGBM), GRU, KF, bidirectional

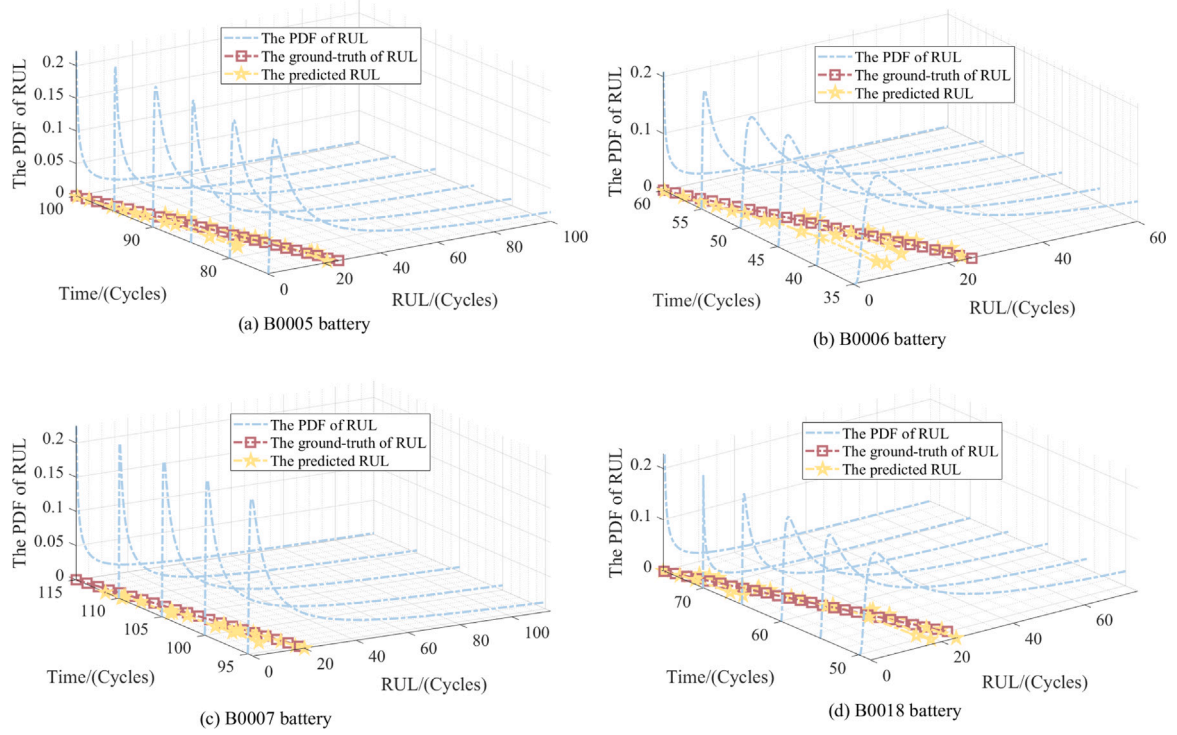


Fig. 4. The RUL probability density functions of the four batteries.

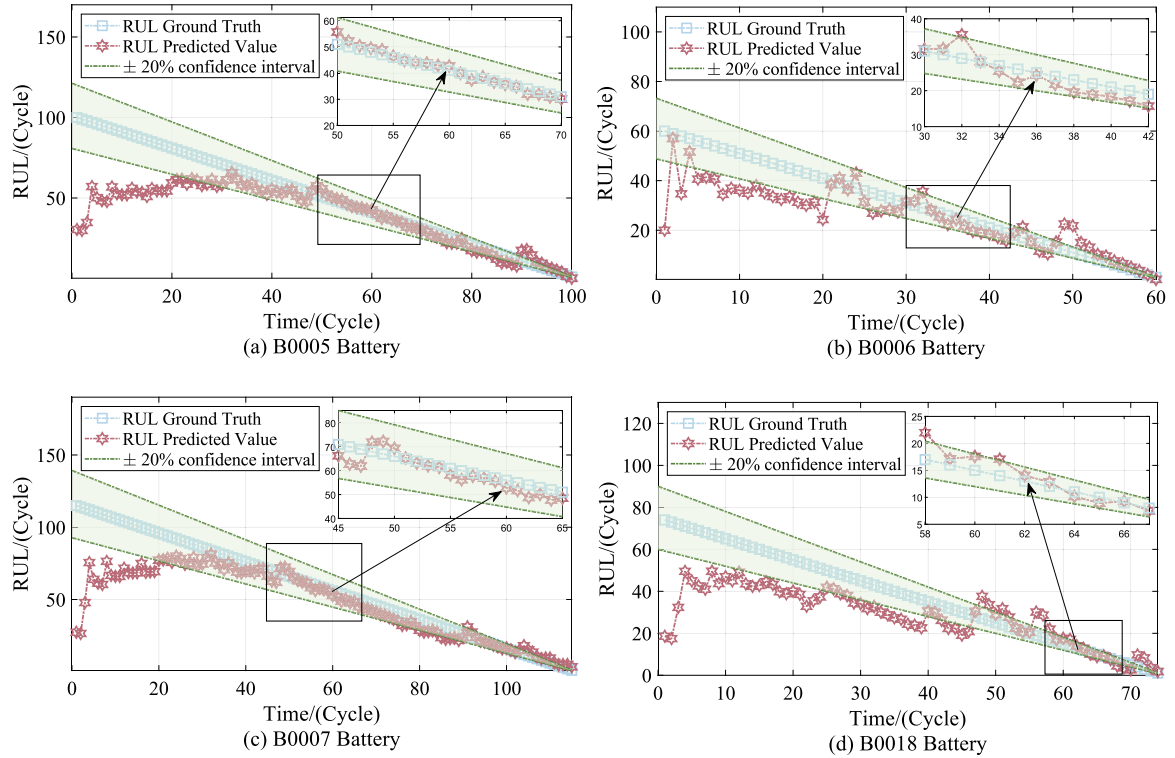


Fig. 5. Experimental results of the RUL prediction of the four batteries.

GRU (Bi-GRU), EKF, LSTM, and bidirectional LSTM (Bi-LSTM). The method of leave-one-out cross-validation is used. Specifically, the data of each battery are used as testing data for online RUL prediction in turn. Alternatively, the data of other batteries are used as offline data to train the model. The evaluation indices, RMSE and MAE, of multiple algorithms for the last 50% working cycles in the battery degrading

process are shown in Table 2. Taking EKF, LSTM, and Bi-LSTM as examples, the average RMSE of each method are 7.075, 7.074 and 6.949, respectively. By contrast, the average RMSE of our proposed approach is 4.354. The decreases in RMSE of the proposed approach are 38.46%, 38.45%, 37.34%, respectively. To better represent the superiority of the proposed approach, Table 3 lists the comparison results

Table 2
Comparison of experimental results of multiple existing approaches.

Approach	RMSE					MAE				
	B0005	B0006	B0007	B0018	Average	B0005	B0006	B0007	B0018	Average
LR	25.529	46.307	30.704	20.928	30.867	21.800	40.100	26.207	18.000	26.527
DTR	9.038	45.503	17.488	11.889	20.980	7.040	43.400	11.897	8.757	17.774
XGBoost	7.765	45.444	16.042	11.056	20.077	6.068	43.486	11.130	8.085	17.192
RFR	6.034	45.063	13.205	9.218	18.380	5.030	43.346	10.000	6.815	16.298
AdaBoost	7.220	44.485	12.743	6.536	17.746	6.400	42.551	10.614	5.594	16.290
PLS	4.053	47.629	9.885	8.848	17.604	3.473	46.338	8.992	7.744	16.637
GBR	5.542	45.163	11.281	7.358	17.336	4.501	43.373	9.314	5.912	15.775
SVR	6.469	43.146	7.758	11.632	17.251	4.760	42.354	7.123	10.447	16.171
KNN	6.685	44.494	10.437	6.536	17.038	5.592	42.973	8.586	5.097	15.562
LightGBM	5.274	43.500	10.316	8.956	17.012	4.336	42.334	9.103	7.779	15.889
GRU	5.505	11.951	6.406	8.205	8.017	4.662	9.782	5.958	6.243	6.661
KF	5.695	7.643	7.392	8.289	7.255	4.718	6.182	6.894	6.457	6.063
Bi-GRU	4.150	9.362	7.489	7.406	7.102	3.523	8.361	6.152	5.379	5.854
EKF	5.408	7.961	7.128	7.801	7.075	4.102	5.596	6.102	5.722	5.381
LSTM	5.439	7.160	6.424	9.273	7.074	4.597	5.750	6.019	6.969	5.834
Bi-LSTM	4.216	9.566	6.401	7.613	6.949	3.607	8.645	5.936	5.865	6.013
Ours	2.844	4.152	3.883	6.535	4.354	2.185	3.080	3.366	5.182	3.453

Table 3
Comparison results of absolute error at the 60th and the 70th working cycles.

Cycle	Approach	Absolute error			
		B0005	B0006	B0018	Average
60	PF	39	27	34	33
	PF-AR	26	24	14	21
	EKPF-AR	18	11	8	12
	Ours	6	8	5	6
70	PF	33	23	18	25
	PF-AR	14	17	5	12
	EKPF-AR	9	7	3	6
	Ours	2	6	1	3

of absolute error at the 60th and the 70th cycles between the proposed approach and some published hybrid empirical approaches in [45]. It should be noted that the same capacity failure threshold is used as [45] for the comparison. It can be learned that, compared with the existing approaches, the proposed adaptive RUL prediction approach can achieve excellent prediction performance under the condition of unlabeled small sample.

4.4. Discussion

Through the comparison in Tables 2 and 3, it can be concluded that the proposed approach can predict RUL well, the prediction performance of which is superior to the machine learning and deep learning approaches. Furthermore, conventional data-driven approaches require a large amount of historical data and the corresponding RUL labels for training to complete the prediction project. Conversely, without any historical data containing labels for training, the RUL prediction approach proposed in this paper is adaptive and unsupervised, which has great advantages for solving the problem of unlabeled small samples. In addition, the distribution of model parameter α_k is integrated in the expression of RUL probability density function, which describes the uncertainty of the parameter estimation. Experimental results in Fig. 4 show that with the accumulation of battery capacity data, the probability density function of the RUL becomes sharper. The reason is that the estimation uncertainty of the parameter α_k in the battery degradation model is decreased, which means that the proposed RUL prediction approach has great adaptability. On the other hand, in order to overcome the constraints of strong Markov characteristics, the estimations of α_k and β_k^2 are processed based on the battery capacity data until the k th working cycle, which is also a contribution of this work. Although there are differences in the working conditions of the four batteries in NASA dataset, it can be seen from Fig. 5 that all the batteries can basically fall within the confidence interval in the later

cycles of the battery operation. In this sense, it can be known that the proposed approach has great robustness.

5. Conclusion

This paper proposes a novel adaptive approach called KF-EM-RTS to predict the battery's RUL in the condition of unlabeled small sample data and uncertainty in the parameter estimation in the battery degradation model. The novelties of the proposed approach include: (1) Without any historical data for offline learning, only the capacity data are needed to accomplish the RUL prediction for a single battery under the situation of unlabeled small sample data. (2) EM-RTS is proposed to adaptively estimate the unknown parameters in the dynamic degradation model, which can overcome the constraints of strong Markov characteristics. (3) The optimal model parameters and the explicit probability density function of the RUL are obtained with solid theoretical foundation. The real-world battery dataset is used to analyze the RUL adaptive prediction approach. The adaptive parameter estimation results are shown in Fig. 3. In addition, the RUL probability density functions in the dataset under several different working cycles during the degradation process are shown in Fig. 4. Experimental results verify that the proposed approach outperforms the existing approaches for predicting the battery's RUL. Compared with EKF, LSTM, and Bi-LSTM, taking average RMSE as an example, the decreased percentages of the proposed approach are 38.46%, 38.45%, 37.34%, respectively. In general, the approach proposed in this paper can accurately predict the RUL of the battery, so as to provide good power performance, and to promote the reliability of the battery in practice.

CRedit authorship contribution statement

Jiushi Zhang: Conceptualization, Methodology, Validation, Writing – original draft. **Yuchen Jiang:** Conceptualization, Methodology, Writing – review & editing. **Xiang Li:** Conceptualization, Methodology. **Mingyi Huo:** Conceptualization, Methodology. **Hao Luo:** Supervision, Writing – review & editing, Funding acquisition. **Shen Yin:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported by National Natural Science Foundation of China under Grant 62073104.

References

- [1] Mishra M, Martinsson J, Rantatalo M, Goebel K. Bayesian hierarchical model-based prognostics for lithium-ion batteries. *Reliab Eng Syst Saf* 2018;172:25–35.
- [2] Lorenzo C, Bouquain D, Higon S, Hissel D. Synthesis of degradation mechanisms and of their impacts on degradation rates on proton-exchange membrane fuel cells and lithium-ion nickel-manganese-cobalt batteries in hybrid transport applications. *Reliab Eng Syst Saf* 2021;107369.
- [3] Patnaik L, Praneeth A, Williamson SS. A closed-loop constant-temperature constant-voltage charging technique to reduce charge time of lithium-ion batteries. *IEEE Trans Ind Electron* 2018;66(2):1059–67.
- [4] Jiang Y, Yin S, Dong J, Kaynak O. A review on soft sensors for monitoring, control and optimization of industrial processes. *IEEE Sens J* 2020.
- [5] Omariba ZB, Zhang L, Sun D. Review of battery cell balancing methodologies for optimizing battery pack performance in electric vehicles. *IEEE Access* 2019;7:129335–52.
- [6] Dubarry M, Devie A. Battery durability and reliability under electric utility grid operations: Representative usage aging and calendar aging. *J Energy Storage* 2018;18:185–95.
- [7] Yu J. State of health prediction of lithium-ion batteries: Multiscale logic regression and Gaussian process regression ensemble. *Reliab Eng Syst Saf* 2018;174:82–95.
- [8] Liu K, Shang Y, Ouyang Q, Widanage WD. A data-driven approach with uncertainty quantification for predicting future capacities and remaining useful life of lithium-ion battery. *IEEE Trans Ind Electron* 2020;68(4):3170–80.
- [9] Zhang J, Jiang Y, Wu S, Li X, Luo H, Yin S. Prediction of remaining useful life based on bidirectional gated recurrent unit with temporal self-attention mechanism. *Reliab Eng Syst Saf* 2022;108297.
- [10] Zhang Y, Peng Z, Guan Y, Wu L. Prognostics of battery cycle life in the early-cycle stage based on hybrid model. *Energy* 2021;221:119901.
- [11] Yin S, Rodriguez-Andina JJ, Jiang Y. Real-time monitoring and control of industrial cyberphysical systems: With integrated plant-wide monitoring and control framework. *IEEE Ind Electron Mag* 2019;13(4):38–47.
- [12] Jiang Y, Yin S, Li K, Luo H, Kaynak O. Industrial applications of digital twins. *Philos. Trans. R. Soc. A* 2021;379(2207):20200360.
- [13] Lipu MH, Hannan M, Hussain A, Hoque M, Ker PJ, Saad MM, Ayob A. A review of state of health and remaining useful life estimation methods for lithium-ion battery in electric vehicles: Challenges and recommendations. *J Clean Prod* 2018;205:115–33.
- [14] Meng H, Li Y-F. A review on prognostics and health management (PHM) methods of lithium-ion batteries. *Renew Sustain Energy Rev* 2019;116:109405.
- [15] Hu X, Che Y, Lin X, Deng Z. Health prognosis for electric vehicle battery packs: A data-driven approach. *IEEE/ASME Trans Mechatronics* 2020;25(6):2622–32.
- [16] Gao D, Huang M. Prediction of remaining useful life of lithium-ion battery based on multi-kernel support vector machine with particle swarm optimization. *J Power Electron* 2017;17(5):1288–97.
- [17] Zhang J, Jiang Y, Luo H, Yin S. Prediction of material removal rate in chemical mechanical polishing via residual convolutional neural network. *Control Eng Pract* 2021;107:104673.
- [18] Shi Z, Chehade A. A dual-LSTM framework combining change point detection and remaining useful life prediction. *Reliab Eng Syst Saf* 2021;205:107257.
- [19] Wang S, Jin S, Bai D, Fan Y, Shi H, Fernandez C. A critical review of improved deep learning methods for the remaining useful life prediction of lithium-ion batteries. *Energy Rep* 2021;7:5562–74.
- [20] Park K, Choi Y, Choi WJ, Ryu H-Y, Kim H. LSTM-based battery remaining useful life prediction with multi-channel charging profiles. *IEEE Access* 2020;8:20786–98.
- [21] Rouhi Ardeshiri R, Ma C. Multivariate gated recurrent unit for battery remaining useful life prediction: A deep learning approach. *Int J Energy Res* 2021.
- [22] Gou B, Xu Y, Feng X. State-of-health estimation and remaining-useful-life prediction for lithium-ion battery using a hybrid data-driven method. *IEEE Trans Veh Technol* 2020;69(10):10854–67.
- [23] Lyu Z, Wang G, Gao R. Li-ion battery prognostic and health management through an indirect hybrid model. *J Energy Storage* 2021;42:102990.
- [24] Li S, Fang H, Shi B. Remaining useful life estimation of lithium-ion battery based on interacting multiple model particle filter and support vector regression. *Reliab Eng Syst Saf* 2021;210:107542.
- [25] Xue Z, Zhang Y, Cheng C, Ma G. Remaining useful life prediction of lithium-ion batteries with adaptive unscented Kalman filter and optimized support vector regression. *Neurocomputing* 2020;376:95–102.
- [26] Yan W, Zhang B, Zhao G, Tang S, Niu G, Wang X. A battery management system with a Lebesgue-sampling-based extended Kalman filter. *IEEE Trans Ind Electron* 2018;66(4):3227–36.
- [27] Ma Y, Chen Y, Zhou X, Chen H. Remaining useful life prediction of lithium-ion battery based on Gauss–Hermite particle filter. *IEEE Trans Control Syst Technol* 2018;27(4):1788–95.
- [28] Qiu X, Wu W, Wang S. Remaining useful life prediction of lithium-ion battery based on improved cuckoo search particle filter and a novel state of charge estimation method. *J Power Sources* 2020;450:227700.
- [29] Aggab T, Avila M, Vignat P, Kratz F. Unifying model-based prognosis with learning-based time-series prediction methods: Application to li-ion battery. *IEEE Syst J* 2021.
- [30] Sun C, Ma M, Zhao Z, Tian S, Yan R, Chen X. Deep transfer learning based on sparse autoencoder for remaining useful life prediction of tool in manufacturing. *IEEE Trans Ind Inform* 2018;15(4):2416–25.
- [31] Mao W, He J, Zuo MJ. Predicting remaining useful life of rolling bearings based on deep feature representation and transfer learning. *IEEE Trans Instrum Meas* 2019;69(4):1594–608.
- [32] Kim S, Choi YY, Kim KJ, Choi J-I. Forecasting state-of-health of lithium-ion batteries using variational long short-term memory with transfer learning. *J Energy Storage* 2021;41:102893.
- [33] Liao L, Köttig F. Review of hybrid prognostics approaches for remaining useful life prediction of engineered systems, and an application to battery life prediction. *IEEE Trans Reliab* 2014;63(1):191–207.
- [34] Zhang M, Kang G, Wu L, Guan Y. A method for capacity prediction of lithium-ion batteries under small sample conditions. *Energy* 2022;238:122094.
- [35] Zhang H, Mo Z, Wang J, Miao Q. Nonlinear-drifted fractional brownian motion with multiple hidden state variables for remaining useful life prediction of lithium-ion batteries. *IEEE Trans Reliab* 2019;69(2):768–80.
- [36] Xu X, Tang S, Yu C, Xie J, Han X, Ouyang M. Remaining useful life prediction of lithium-ion batteries based on wiener process under time-varying temperature condition. *Reliab Eng Syst Saf* 2021;214:107675.
- [37] Zhou D, Zheng W, Chen S, Fu P, Zhu H, Song B, Qu X, Wang T. Research on state of health prediction model for lithium batteries based on actual diverse data. *Energy* 2021;230:120851.
- [38] Severson KA, Attia PM, Jin N, Perkins N, Jiang B, Yang Z, Chen MH, Aykol M, Herring PK, Fraggadakis D, et al. Data-driven prediction of battery cycle life before capacity degradation. *Nature Energy* 2019;4(5):383–91.
- [39] Lee D, Choi D. Analysis of the reliability of a starter-generator using a dynamic Bayesian network. *Reliab Eng Syst Saf* 2020;195:106628.
- [40] Huang Y, Zhang Y, Li N, Wu Z, Chambers JA. A novel robust student's t-based Kalman filter. *IEEE Trans Aerosp Electron Syst* 2017;53(3):1545–54.
- [41] Huang Y, Zhang Y, Zhao Y, Mihaylova L, Chambers JA. Robust Rauch–Tung–Striebel smoothing framework for heavy-tailed and/or skew noises. *IEEE Trans Aerosp Electron Syst* 2019;56(1):415–41.
- [42] Gibson S, Wills A, Ninness B. Maximum-likelihood parameter estimation of bilinear systems. *IEEE Trans Autom Control* 2005;50(10):1581–96.
- [43] Folks JL, Chhikara RS. The inverse Gaussian distribution and its statistical application—a review. *J. R. Stat. Soc. Ser. B Stat. Methodol.* 1978;40(3):263–75.
- [44] Saha B, Goebel K. Battery data set. In: NASA ames prognostics data repository. Moffett Field, CA: NASA Ames Research Center; 2007. <http://ti.arc.nasa.gov/project/prognostic-data-repository>.
- [45] Zhang N, Xu A, Wang K, Han X, Hong W, Hong SH. Remaining useful life prediction of lithium batteries based on extended Kalman particle filter. *IEEJ Trans Electr Electron Eng* 2021;16(2):206–14.